

Innovation Institute for Tomorrow

Executive Summary

Client

Tim Heffernan

Student Consulting Team

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Background

II4T is an education nonprofit that works to provide students with access to world class STEM education for 21st century job markets. To accomplish this goal they have partnered with K-12 schools, colleges and universities, and businesses and industries to bring advanced STEM education to students across Pennsylvania.

Project Description

Project Opportunity

The main problem our project addressed was that II4T was looking to expand their RAD (Robotics, Artificial Intelligence, Drone Technology) to include an Artificial Intelligence (AI) portion. While II4T already specializes in Robotics and Drone programs for students, they were looking to incorporate a new AI component to further align with their mission of providing students in Western PA exposure to sustainable, world-class STEM education and technology such that they are better prepared the 21st century job market, which now more than ever prioritize candidates with AI skills. Prior to our project, no attempts were made to develop an AI portion of the RAD curriculum, but again considering the mission of II4T, to prepare students for 21st century jobs, current industry trends informed their decision to request help in developing the AI portion of the program.

Project Vision

Our team worked closely with our client to iterate on a project vision during the first few weeks of the semester. Eventually, our team landed on the vision of creating in-house AI curriculum modules, tailored to the education background and skillset of the students in the program. Specifically, our vision was to fully build out 2 modules, complete with content slide decks and activities (where students would use the DARE platform) and then create documentation that details how II4T can maintain the program structure/activities.

Project Outcomes

As for our project outcomes, our team was able to deliver our project vision and also pilot our modules for a student group. On April 14th and April 16th, students completed Module 1 of the AI curriculum asynchronously, working with a “Day 1” content slide deck (covering AI essentials) on April 14th and a “Day 2” content slide deck (covering Real-World Applications of AI) on April 16th and completing the Module activities on both days. On April 18th, our team was able to visit the client on-site in Franklin, PA to teach students Module 2 of the curriculum (covering Prompt Engineering), live instructing students on the actual content of the slide deck as well as helping them progress through the activities in Module 2. Our team

has also compiled all of the content that we created over the semester, including the slide decks, activity instructions, and Google Forms for students to complete as they progress through content, into a Google Drive Folder for the II4T team to reference in the future.

Project Deliverables

Our project deliverables include a pre-content questionnaire for II4T to test the students' baseline knowledge prior to the program. We also have created three slide decks to cover a variety of topics in the field of AI. Each of these slide decks is accompanied by google forms for students to submit their work along the way and answer questions and instruction handouts for curated activities we have planned. Lastly, we have created a post-content questionnaire to assess the students' learning from the program.

Recommendations

To help II4T sustain and build on the progress made during this cohort of students, we identified 2 recommendations. The first is for adding more activities throughout the modules, particularly in the later half of Module 2, and ensuring that all existing activities include a submission component so instructors can verify participation and gauge student understanding throughout each section. The second recommendation is for II4T to consistently administer the pre and post content questionnaires with every future cohort and track results over time, as cross cohort data is far more valuable to funders than a single cohort's results.

Student Consulting Team

Muskan served as project manager. She is a third-year student majoring in Information Systems and Computer Science with a minor in Machine Learning. She will be interning at Roblox this summer as a software engineer.

Martin was the client relationship manager. He is a third-year student majoring in Information Systems and Artificial Intelligence. He will be working for The Children's Home of Pittsburgh this summer as a software developer.

Aditi led quality assurance. She is a third-year student majoring in Information Systems and Human-Computer Interaction. She will be interning at PayPal this summer as a Technical Product Manager.